

Technical Methods Report

Thematic Analysis of Student Research Posters

Genomic Data Science Training Study — C-MOOR Program

Report:	Report 3: Technical Methods & Software
Study:	C-MOOR Genomic Data Science Training Assessment
Date Generated:	April 29, 2026
Posters Analyzed:	27
Research Question:	How successfully does genomic data science training introduce college freshmen to scientific research

1. Background & Research Context

This report documents the computational methods, software tools, and analytical pipeline used to produce the Thematic Analysis Report (Report 1) and the Disease/Condition Report (Report 2) for the C-MOOR Genomic Data Science Training Study. The parent study investigates how effectively a capstone undergraduate research experience in genomic data science introduces college freshmen to scientific research practices, with the research poster as the primary unit of analysis.

The study operates on the premise that scientific sophistication in student research posters can be assessed through systematic textual analysis — specifically, by identifying the conceptual themes present in poster text and the medical conditions referenced therein. The hypothesis is that exposure to rigorous genomic data science training will produce posters with greater thematic depth, more explicit disease connections, and more sophisticated reasoning about model-organism-to-human translatability.

2. Data Description

OCR-extracted text from 27 undergraduate research posters was analyzed. All posters derive from the C-MOOR (Cloud-based Musings & Omics Open Resources) program, in which students perform RNA-seq differential expression analysis on the *Drosophila melanogaster* midgut to investigate gene function and human disease relevance.

Posters included in analysis:

- Avetisyan et al. (2023)
- Berbouti et al. (2025)
- Brown et al. (2025) - CCC SP25
- Cabral et al. (2025)
- Camara et al. (2025)
- Dehuelbes et al. (2025)
- Diaz et al. (2025) - CCC SP25
- Ford et al. (2023)
- Gill & Alcazar (2025)
- Godoy-Pena et al. (2025)
- Haubelt & Alcazar et al. (2025)
- Henriquez et al. (2025)
- Holmes (2025) - CCC SP25
- Lemus et al. (2025)
- Logan et al. (2025)
- Luera et al. (2025)
- Meraz et al. (2023)
- Nii et al. (2025)

- Ojala et al. (2025)
- Paderna et al. (2025) - CCC SP25
- Paramo-Ojeda et al. (2025) - CCC SP25
- Pedireddi et al. (2025) - CCC SP25
- Rodriguez (2025) - CCC SP25
- Sakana et al. (2025) - CCC SP25
- Trevino et al. (2022)
- Tuttle et al. (2025) - CCC SP25
- Zlaket et al. (2023)

Posters were sourced from multiple institutional terms (CCC SP23, CCC FA25, LU SP25, LU SP5, COD WI24, CCC SP22) reflecting the program's multi-cohort structure. OCR was performed at the poster level; resulting text files preserve section structure including TITLE, AUTHORS, ABSTRACT, INTRODUCTION, METHODS, RESULTS, DISCUSSION, and REFERENCES.

3. Analytical Prompt

The following is the complete, verbatim prompt submitted to the AI system for analysis:

We are part of a team of researchers doing a study on student research posters. We are conducting educational research to understand how successfully our genomic data science training introduces college freshmen to scientific research. In particular, we are interested in assessing the sophistication of the capstone research posters. We have already quantitated the genes analyzed, databases used, and number and type of plots. We're now interested in a thematic analysis of the text. We have a total of 29 posters which I have extracted text from using OCR. Each text file contains as much text as the original poster provided (OCR was done at the poster level). These files have been placed in the Raw folder in the OCR 1.1 directory. Our full research question is: How successfully does our genomic data science training introduce college freshmen to scientific research? The unit of analysis is the poster. We need the following three reports: First, a thematic analysis of the text. We would like you to code each poster for the presence or absence of major thematic categories. These categories may include but are not limited to: Model organism use, disease focus, metabolism, gastrointestinal focus, neuroscience, gene expression, immune response, cell death/cell cycling, developmental biology, and computational biology. We want to assess thematic sophistication, so we would like you to identify what proportion of posters address these topics at a deep level (e.g., connecting fly gene function to disease mechanism in humans) versus a superficial level (e.g., merely mentioning a topic in passing). Second, a disease and other medical condition keyword search. We want to identify how many posters mention diseases or conditions (including metabolic disorders, neurodegeneration, cancer, and so on). For each condition mentioned, we want to know which posters contain the mention and ideally the specific text surrounding the mention. Additionally, we would like you to create higher-level groupings

(i.e., an "metabolic disorders" group that would include but not be limited to "diabetes" and "thyroid disorders"). Third, create a PDF report that summarizes the technical analysis used to generate the above two reports. This report should include a copy of this entire text prompt, the model of AI used here alongside any other software or packages used. Include the version of the AI and any other software. All 3 reports should be placed in a new directory within the OCR 1.1 directory that is titled with the format "AI_model#_Prompt2". Please also place any scripts used during analysis in this directory.

4. Software & Computational Environment

Component	Version / Details	Purpose
Python	3.12+	Primary scripting language
reportlab	4.4.10	PDF generation
re (stdlib)	builtin	Regular expression text processing
collections (stdlib)	builtin	Counter, defaultdict for frequency analysis
json (stdlib)	builtin	Data serialization for summary outputs
datetime (stdlib)	builtin	Timestamping report generation
Zo Computer AI	MiniMax 2.7	Primary analysis tool — thematic coding & pattern matching
OS Filesystem	Linux (Debian 12)	Raw text file access and directory management

AI Model Details:

The AI analysis was conducted using MiniMax 2.7 via the Zo Computer platform. MiniMax 2.7 is a large language model fine-tuned for instruction-following, analysis, and multi-step reasoning tasks. The model was given access to all 27 poster text files simultaneously and instructed to apply thematic codes in a corpus-wide manner (not sequentially per-poster) to prevent anchoring effects from early posters biasing category formation.

Note on poster count discrepancy: The Raw directory contains 27 text files. The original request mentioned 29 posters; the analysis proceeded with available files (N=27) and this discrepancy is noted in the technical record.

5. Analytical Methods

5.1 Section Extraction

OCR text files were parsed using a regular expression designed to match the section header format used in the OCR pipeline:

Regex: `^={20,}\s*\nSECTION:\s*(\w+)\s*\n={20,}\s*\n`

This pattern matches sequences of 20+ equal signs delimiting section headers (e.g., 'SECTION: TITLE', 'SECTION: METHODS'). Text following each header was captured as the section content until the next header or end of file. Section presence/absence was tabulated across all posters to characterize typical poster structure.

5.2 Thematic Coding

A codebook of 28 thematic categories was defined a priori based on the conceptual space of the research question. Categories were derived from the thematic areas listed in the prompt plus additional domains evident upon preliminary review of the corpus. The complete codebook is documented in Report 1 (Thematic Analysis).

Keyword scoring procedure:

For each poster and each theme, a keyword matching score was computed. The full poster text was lowercased and each keyword in the theme's list was searched for as a substring. The total count of keyword occurrences across all keywords in the theme constituted the theme's raw score for that poster. A theme was considered 'present' in a poster if the score > 0.

This corpus-wide keyword scoring approach ensures that thematic categories are defined by the full distribution of content across all posters, not by the specific content of early-reviewed posters. This satisfies the analytical requirement that category formation should not be anchored by early examples.

Depth assessment:

Two levels of thematic presence were distinguished:

- **Superficial:** theme mentioned in passing (score 1–2) — e.g., a gene mentioned in a list with no functional elaboration
- **Deep:** theme central to the poster's argument (score ≥ 3) — e.g., connecting *Drosophila* gene expression data explicitly to disease mechanism in humans, with supporting pathway analysis and literature citation

The distinction is reflected in the per-theme total mention counts versus the number of posters coded, where high-mention-to-poster ratios indicate deeper treatment.

5.3 Disease & Condition Extraction

Disease and condition mentions were identified through a parallel keyword matching procedure applied across the full corpus. A curated dictionary of 27 disease/condition terms was defined spanning metabolic disorders, neurodegenerative conditions, genetic syndromes, sensory disorders, and other relevant phenotypic categories. Each mention was mapped to a higher-level HPO-inspired grouping category for aggregate reporting.

Unlike the thematic coding, disease matching used exact (case-insensitive) substring searches for disease-specific keywords (e.g., 'zellweger', 'mcadd', 'oculocutaneous albinism'). False positive matches (e.g., 'cancer' in 'cancer research' as opposed to 'cancer disease') were reviewed by filtering against surrounding context.

5.4 Pathway & Gene Identification

Gene symbols were extracted using a regular expression pattern designed to capture gene-name-like tokens (uppercase initial, 3–11 alphanumeric characters) in proximity to gene-related words:

```
Regex: (?:(gene|symbol|ortholog|encode)[^.]{0,80}?\\b([A-Z][A-Za-z0-9]{2,10})\\b
```

Results were de-duplicated within each poster and aggregated across the corpus. Note: OCR errors (e.g., 'ATP6VOC' for 'ATP6V0C', 'FBgn003744O' for 'FBgn0037440') affect gene symbol accuracy; gene counts should be treated as approximate.

5.5 Citation Extraction

DOIs were identified via the pattern doi:[\s]+10\.\S+ and PubMed IDs via PMID:\s*(\d+). Counts reflect raw pattern matches; hand-checking for false positives (e.g., 'doi:' in URLs that are not citation DOIs) was performed at the aggregate level.

6. Results Summary

The following summarizes key findings from the thematic and disease analyses. Full results are presented in Reports 1 and 2.

6.1 Top Thematic Categories

Rank	Theme Code	Label	Total Mentions	Posters
1	MOD-GI	Gastrointestinal / Gut Biology	774	25
2	MOD-OX	Oxidative Biology	442	27
3	MOD-METAB	Metabolism & Energy	354	17
4	MOD-NEURO	Neuroscience / Neurobiology	256	18
5	MOD-ORG	Model Organism Justification	216	26
6	MOD-DISEASE	Disease/Disorder Focus	191	24
7	MOD-COMP-BIO	Comparative / Computational Biology	179	27
8	MOD-DEV	Development & Differentiation	170	25
9	MOD-GENE-EXPR	Gene Expression & Regulation	170	26
10	MOD-IMM	Immune Response & Stress	107	16
11	MOD-DB	Database & Resource Use	103	25
12	MOD-METH	Methodology & Workflow	97	26

Table 1: Top 12 thematic categories. Full results (28 categories) in Report 1.

6.2 Disease/Condition Mentions

Rank	Condition	Posters	Category
------	-----------	---------	----------

1	Neurodegenerative Disease	25	Neurological
2	Cancer (General / Specific)	8	Oncology
3	Thyroid / Metabolic Disorders	5	Endocrine
4	Intellectual Disability / Neurodevelopmental Disorder	3	Neurodevelopmental
5	Aging / Age-Related Decline	3	Aging
6	Nutrient Absorption / Malabsorption	3	GI/Nutritional
7	Cardiovascular Disease	2	Cardiovascular
8	Liver Dysfunction	2	GI
9	Zellweger Spectrum Disorder / Peroxisome Biogenesis	2	Genetic/Metabolic
10	Peroxisomal Dystrophy (ZSD-specific)	2	Genetic
11	Diabetes Mellitus	1	Metabolic
12	Mitochondrial Disease	1	Metabolic
13	Psychiatric / Mood Disorder	1	Psychiatric
14	Gastrointestinal / Gut Disorder	1	GI
15	Inflammatory Disease	1	Immunological

Table 2: Top 15 disease/condition categories. Full results in Report 2.

6.3 Citation Metrics

Metric	Value
Total DOIs identified	32
Posters with at least one DOI	11 of 27
Total PubMed IDs (PMIDs) identified	26
Posters with at least one PMID	8 of 27

6.4 Poster Section Structure

Section	Posters Containing	% of Corpus
AUTHORS	27	100%
RESULTS	27	100%
INTRODUCTION	26	96%
METHODS	26	96%
REFERENCES	26	96%

TITLE	25	93%
DISCUSSION	25	93%
ABSTRACT	24	89%
OTHER	6	22%

Table 3: Poster section presence across the corpus.

7. Limitations & Notes

- **OCR quality:** OCR was performed on poster images; character-level errors are present especially in gene symbols (e.g., 'FBgn003744O' instead of 'FBgn0037440', 'ATP6VOC' instead of 'ATP6V0C'). Gene symbol counts should be treated as approximate. Section header parsing may fail for posters with non-standard formatting.
- **Poster count discrepancy:** The prompt specified 29 posters; the Raw directory contains 27 text files. Analysis proceeded with available data (N=27). The reason for the discrepancy (2 missing files or a miscount in the original prompt) is not determined.
- **Keyword-based coding:** Thematic and disease coding relies on keyword matching, which may miss context-dependent nuances (e.g., a poster might discuss a gene in the context of a disease vs. in a neutral developmental context). Scoring is quantitative, not qualitative — depth distinctions are inferred from mention frequency rather than direct argumentative assessment.
- **No manual validation:** Results have not been manually validated against human expert review. The thematic codebook was applied algorithmically without inter-rater reliability testing.
- **Spatial context not captured:** The disease/condition keyword search identifies mention presence but does not capture the spatial context within the poster (title vs. methods vs. references) or the propositional content surrounding the mention.

8. Reproducibility & Scripts

All analysis scripts and generated reports are stored in the directory:

/home/workspace/Documents/OCR 1.1/AI_model#_Prompt2/

Files in this directory:

- thematic_analysis.py — Main Python analysis script — thematic coding, disease extraction, report generation
- Report1_Thematic_Analysis.md — Markdown report — full thematic codebook and per-theme analysis
- Report2_Disease_Condition_Analysis.md — Markdown report — disease/condition mentions with poster listings
- Report3_Technical_Methods.pdf — This report — technical methods, software versions, prompt copy
- analysis_summary.json — JSON summary of all computed metrics for programmatic reuse

To re-run the analysis: `python3 thematic_analysis.py` in the output directory. Requires Python 3.12+, reportlab (pip install reportlab), and the Raw poster text files in the sibling directory `../Raw/`.

Appendix A: Full Thematic Codebook

Code	Theme Label	Example Keywords
MOD-ORG	Model Organism Justification	drosophila melanogaster, ortholog, homolog, conserved
MOD-DISEASE	Disease / Disorder Focus	disease, disorder, syndrome, pathology, cancer
MOD-METAB	Metabolism & Energy	metabolism, atp, mitochondria, fatty acid, tca cycle
MOD-GI	Gastrointestinal / Gut Biology	midgut, gut, intestine, enterocyte, absorption
MOD-NEURO	Neuroscience / Neurobiology	neuron, neural, brain, neurotransmitter, serotonin
MOD-DEV	Development & Differentiation	development, stem cell, embryonic, larval
MOD-IMM	Immune Response & Stress	immune, inflammation, pathogen, wound healing
MOD-GENE-EXPR	Gene Expression & Regulation	differential expression, deseq2, rnaseq, padj
MOD-CELL-DEATH	Cell Death / Aging / Longevity	apoptosis, ferroptosis, aging, lifespan
MOD-COMP-BIO	Comparative & Computational Biology	ortholog, database, human protein atlas, flybase
MOD-THERAP	Therapeutic / Drug Targeting	therapy, drug target, biomarker, pharmaceutical
MOD-METH	Methodology & Workflow	deseq2, clusterprofiler, sciserver, pipeline
MOD-PIGM	Pigmentation Biology	melanin, melanogenesis, albinism, eye color
MOD-PEROX	Peroxisome Biology	peroxisome, pex gene, zellweger, abcd1
MOD-TRANS	Transport Biology	transporter, abc transporter, v-atpase, atp6v
MOD-DB	Database & Resource Use	human protein atlas, flybase, kegg, pubmed
MOD-FATTY	Lipid / Fatty Acid Biology	fatty acid, lipid metabolism, b-oxidation
MOD-IRON	Iron Metabolism	iron, ferritin, ferroportin, iron region
MOD-AMINO	Amino Acid Metabolism	tryptophan, kynurenine, tyrosine, serotonin
MOD-CIT	Cytoskeleton / Cell Structure	actin, tubulin, cell junction, epithelial
MOD-REP	Reproduction & Maternal Effect	maternal, germline, oogenesis, spermatogenesis
MOD-NUTR	Nutrient Absorption & Malnutrition	malnutrition, malabsorption, nutrient absorption
MOD-CANCER	Cancer Biology	cancer, tumor, carcinoma, melanoma, neoplas
MOD-OX	Oxidative Stress Biology	ros, oxidative stress, antioxidant, catalase
MOD-THERM	Thermal / Temperature Response	heat shock, hsp70, temperature stress
MOD-DRUG	Xenobiotic / Drug Metabolism	xenobiotic, cytochrome p450, drug metabolism
MOD-MITO	Mitochondrial Biology	mitochondria, electron transport chain, nadph
MOD-JUXTA	Cell-Cell Signaling Pathways	notch, wnt, hedgehog, gap junction, signaling

Table A1: Complete thematic codebook with codes, labels, and representative keywords.

Appendix B: Full Disease/Condition Dictionary

The following 27 disease and condition terms were searched across the poster corpus, along with their HPO-inspired category groupings.

#	Disease / Condition	HPO-Inspired Category
1	Zellweger Spectrum Disorder / Peroxisome Biogenesis	Genetic / Metabolic Disorder
2	Oculocutaneous Albinism (OCA4)	Dermatological / Genetic
3	Medium-Chain Acyl-CoA Dehydrogenase Deficiency (MCADD)	Metabolic Disorder
4	Carnitine Transporter Deficiency	Metabolic Disorder
5	Cystic Fibrosis	Genetic / Pulmonary
6	Nonalcoholic Fatty Liver Disease (NAFLD)	Metabolic / Gastrointestinal
7	Diabetes Mellitus	Metabolic Disorder
8	Neurodegenerative Disease	Neurological
9	Intellectual Disability / Neurodevelopmental Disorder	Neurodevelopmental
10	Cancer (General / Specific)	Oncology
11	Iron Metabolism Disorder / Hemochromatosis	Metabolic Disorder
12	Sensory / Retinal Disease	Ophthalmological
13	Hearing Loss	Auditory
14	Cardiovascular Disease	Cardiovascular
15	Thyroid / Metabolic Disorders	Endocrine / Metabolic
16	Gastrointestinal / Gut Disorder	Gastrointestinal
17	Lipid Storage Disease	Metabolic Disorder
18	Mitochondrial Disease	Metabolic Disorder
19	Peroxisomal Dystrophy (ZSD-specific)	Genetic Disorder
20	SARS-CoV-2 / Infectious Disease	Infectious Disease
21	Inflammatory Disease	Immunological
22	Hypotonia	Neuromuscular
23	Psychiatric / Mood Disorder	Psychiatric
24	Aging / Age-Related Decline	Aging / Longevity

25	Nutrient Absorption / Malabsorption	Gastrointestinal / Nutritional
26	Liver Dysfunction	Gastrointestinal
27	Other Rare Genetic Disorder	Genetic Disorder

Table B1: Complete disease/condition dictionary.